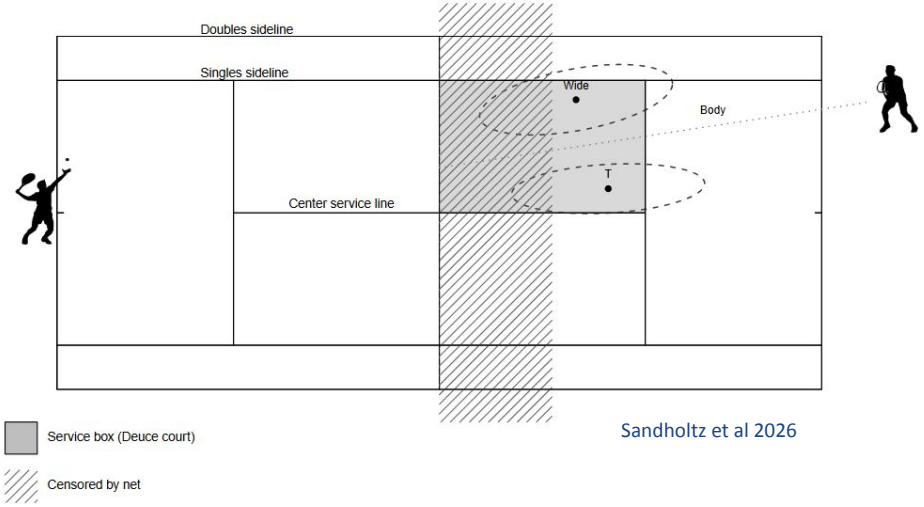
Optimal Serve Placement in Professional Tennis Accounting for Execution Error

Brett Pedersen and Nathan Sandholtz



Goals

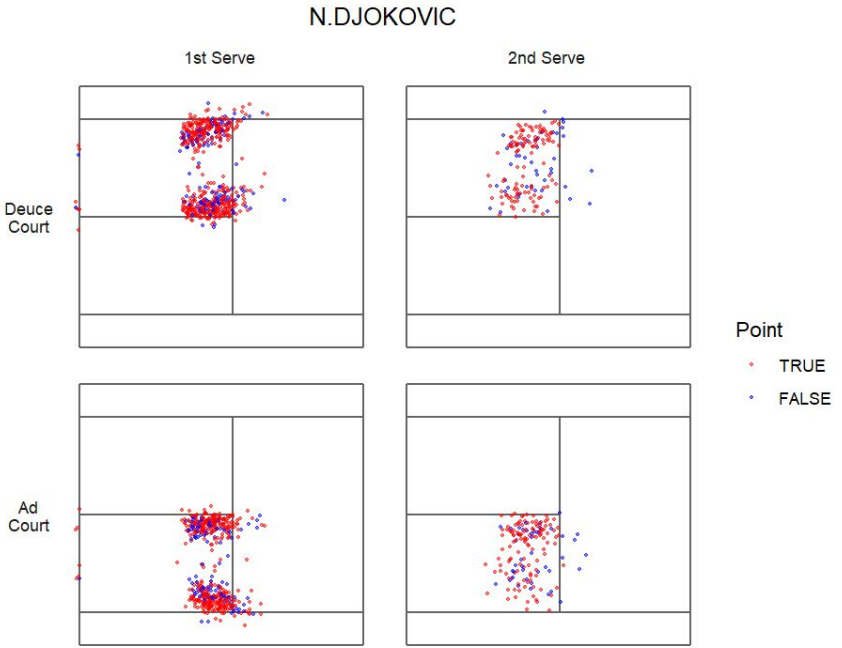
- Quantify the relationship between serve landing location and point win probability (Reward Surface)
- Estimate each server's intended targets and precision level (Execution Error)
- Find optimal targets according to the methodology outlined in Sandholtz et al 2026 (Optimization)
- Show that aiming closer to our prescribed optima is associated with winning more points (Validation)



Sandholtz et al 2026

Data

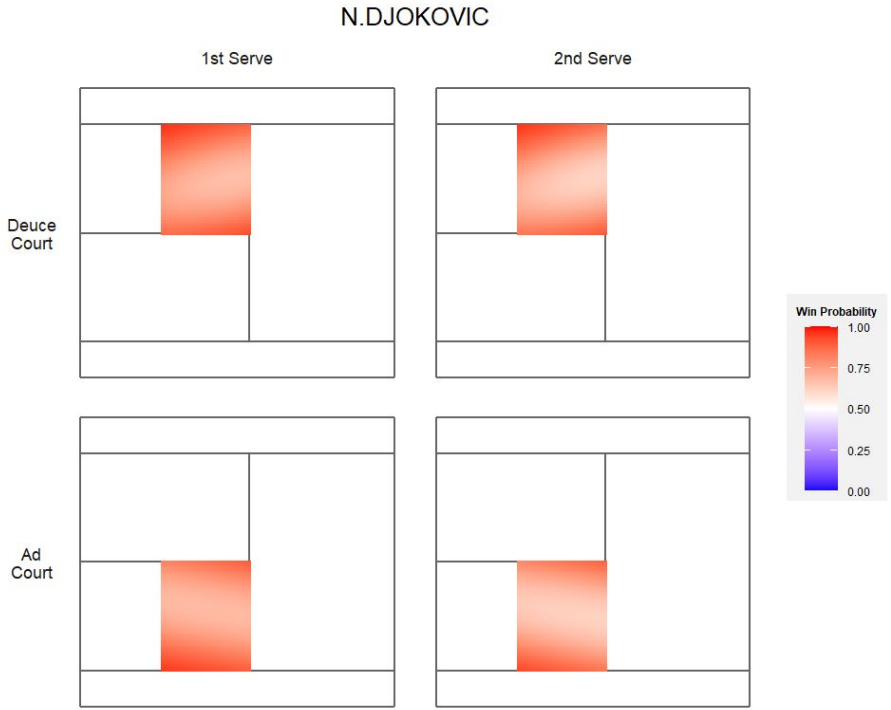
- Spatial serve data from the 2019-2021 Australian Open professional tennis tournaments
- (x, y) bounce locations, point winner, server, returner, serve number, court-side, and other contextual variables
- Point outcomes enable validation of prescribed optimums (an advantage over the BYU collegiate experiment)



Reward Surface

Goal: Quantify the relationship between serve landing location and point win probability

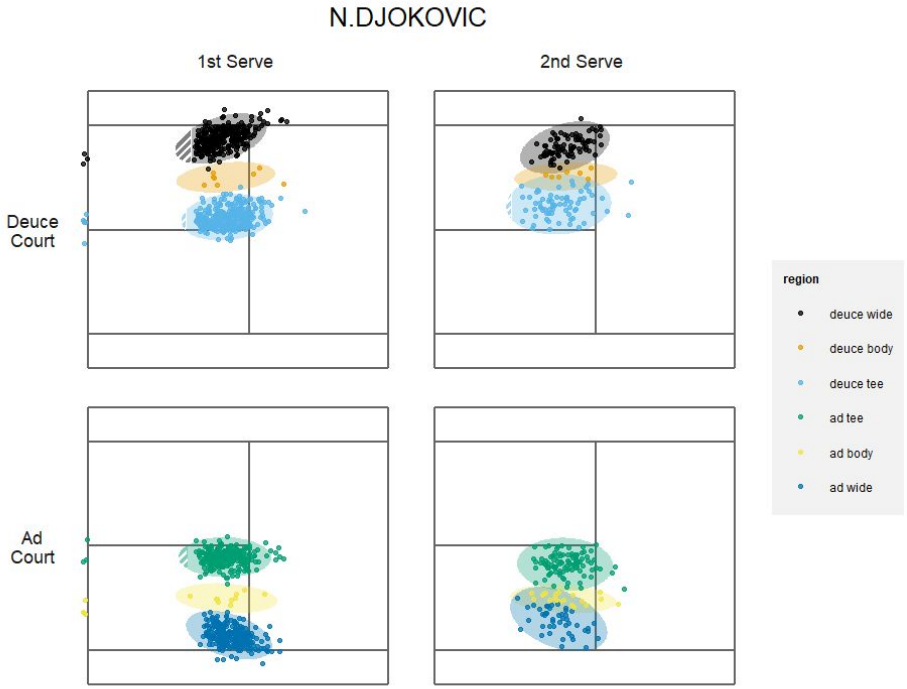
- Generalized Additive Mixed Model (GAMM)
- For each serve, whether the server won the point is treated as a binary response
- Fit a smooth surface over serve landing location and speed
- Similar surfaces for Deuce-court/Right-handed serves and Ad-court/Left-handed serves
- Include server and returner identifiers as random intercepts



Execution Error

Goal: Estimate each server's intended targets and precision level

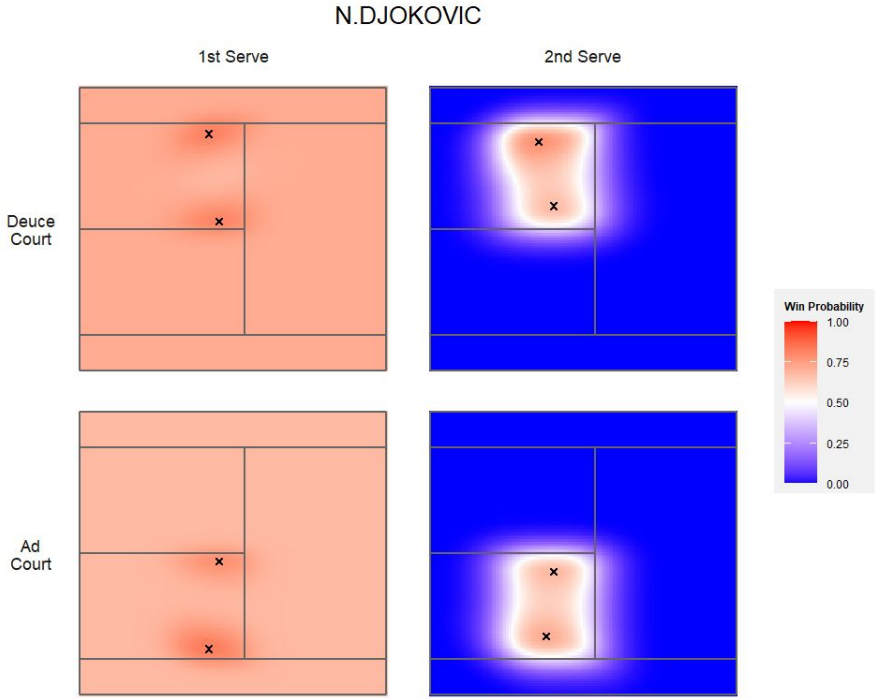
- Bayesian Bivariate Gaussian Mixture Model (GMM)
- For each combination of server, serve number, and court-side, serves follow a 3-component GMM
- Left censoring bounds that are induced by the net
- Hierarchical pooling between players



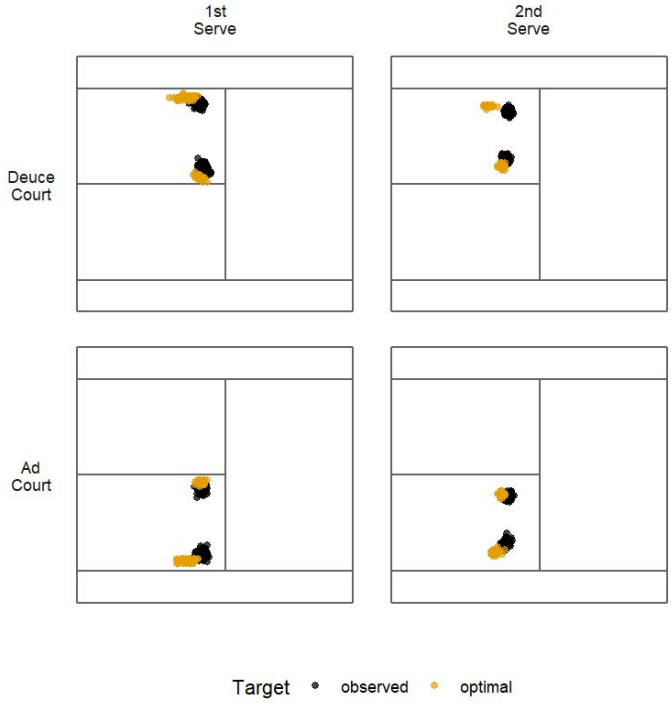
Optimization

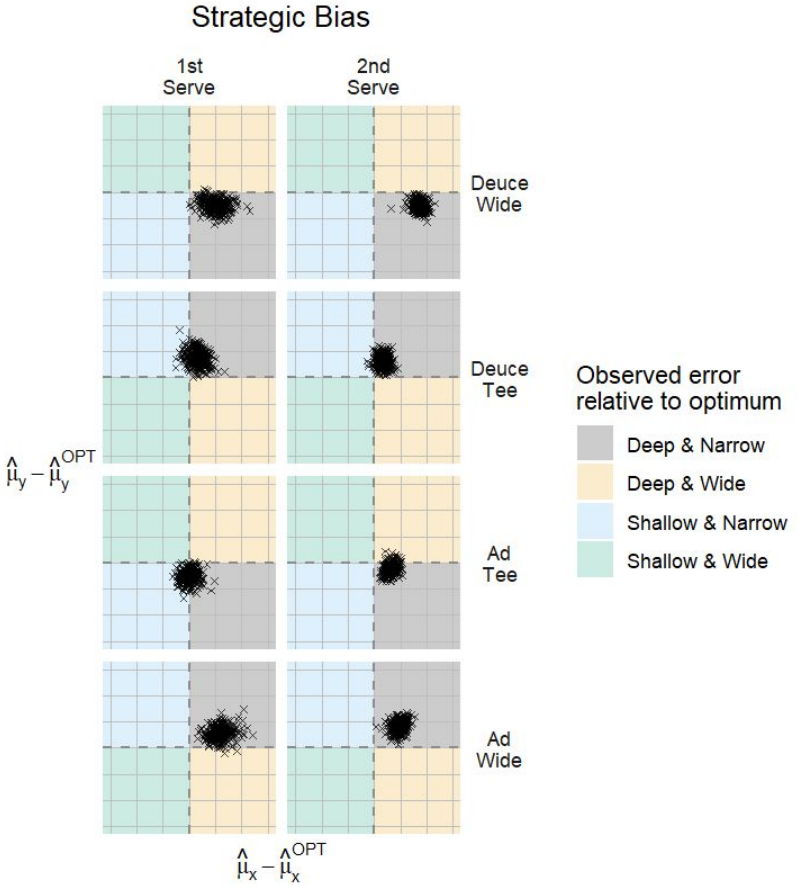
Goal: Find optimal targets according to the methodology outlined in Sandholtz et al 2026

- Treating the tennis serve as a 2-period Markov Decision Process (MDP)
- For a potential target, define a bivariate gaussian distribution of serves aimed there (requires interpolation)
- Weight point outcomes from the reward surface according to the densities of the newly defined serve distribution
- Identify local maxima



Bias Trends


1st Serve
2nd Serve
Deuce Court
Ad Court
Target
observed
optimal


Strategic Bias
1st Serve
2nd Serve
Deuce Wide
Deuce Tee
Ad Tee
Ad Wide
$\hat{\mu}_y - \hat{\mu}_y^{OPT}$
$\hat{\mu}_x - \hat{\mu}_x^{OPT}$
Observed error relative to optimum
Deep & Narrow
Deep & Wide
Shallow & Narrow
Shallow & Wide

Validation

Goal: Show that aiming closer to our prescribed optima is associated with winning more points

- Modeling the effect of distance (between intended targets and optimal targets) on point win probability for each serve number, court-side, and strategy
- Generalized Linear Hypothesis Test

Serve	Court Side	Direction (Strategy)	Estimate	Std. Error	z	p
1st	Ad	T	−0.315	0.154	−2.05	0.121
	Ad	Wide	−0.321	0.104	−3.10	0.007**
	Deuce	T	−0.514	0.124	−4.15	<0.001***
	Deuce	Wide	−0.195	0.092	−2.13	0.101
2nd	Ad	T	−0.090	0.172	−0.52	0.839
	Ad	Wide	−0.097	0.116	−0.83	0.7
	Deuce	T	−0.289	0.126	−2.29	0.069
	Deuce	Wide	0.029	0.082	0.36	0.992

References

Stephanie Kovalchik, Martin Ingram, Kokum Weeratunga, and Cagatay Goncu. Space-time von cramm: Evaluating decision-making in tennis with variational generation of complete resolution arcs via mixture modeling. arXiv preprint arXiv:2005.12853, 2020.

Matt Ludwig, Ryan S. Brill, and Abraham J. Wyner. Separating intent from execution: A probabilistic approach to pitch location accuracy, 2025. URL https://arxiv.org/abs/2508.19184.

Nathan Sandholtz, Ryan Hanson, Ron Hager, Stephanie Kovalchik, and Gilbert Fellingham. Stated, realized, and optimal aiming strategies for the tennis serve: Experimental evidence from collegiate athletes. arXiv preprint arXiv:2607.28500, 2026.

Mark Walker and John Wooders. Minimax play at wimbledon. American Economic Review, 91(5): 1521–1538, 2001.

Questions?

Reward Surface

$$Point_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha + \beta \cdot \mathbf{1}[Hand\text{-}Court_i = \text{"Cross Body"}] + f_{Hand\text{-}Court_i}(x_i, y_i, speed_i) + u_{\text{server}[i]} + u_{\text{returner}[i]}$$

$$u_{\text{server}[i]} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_{\text{server}})$$

$$u_{\text{returner}[i]} \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_{\text{returner}})$$

Execution Error

$$f(z_i) = \sum_{d=1}^{3} \theta_{p,s,c,d} \cdot \mathcal{N}_2\left(z_i \mid \boldsymbol{\mu}_{p,s,c,d},\, \Sigma_{p,s,c,d}\right)$$

$$\Sigma_{p,s,c,d} = \begin{pmatrix} \tau^x_{p,s,c,d} & 0 \\ \rho_{p,s,c,d}\,\tau^y_{p,s,c,d} & \sqrt{1-\rho^2_{p,s,c,d}}\,\tau^y_{p,s,c,d} \end{pmatrix} \begin{pmatrix} \tau^x_{p,s,c,d} & 0 \\ \rho_{p,s,c,d}\,\tau^y_{p,s,c,d} & \sqrt{1-\rho^2_{p,s,c,d}}\,\tau^y_{p,s,c,d} \end{pmatrix}^\top$$

$$p(\mathrm{net}_j \mid p, s, c) = \sum_{d=1}^{3} \theta_{p,s,c,d} \cdot \Phi\left(t_{p,s,c,d} \mid \mu^x_{p,s,c,d},\, \tau^x_{p,s,c,d}\right)$$

Execution Error

Location

$$\mu^x_{p,s,c,d} \sim \mathcal{N}\left(\mu^x_{g,s,c,d},\, \sigma_\mu\right), \quad \mu^x_{g,s,c,d} \sim \mathcal{N}(6,\, 1.5), \quad \mu^y_{p,s,c,d} \sim \mathcal{N}\left(\mu^y_{g,s,c,d},\, \sigma_\mu\right), \quad \mu^y_{g,s,c,d} \sim \mathcal{N}(0,\, 2.5)$$

$$\mu^y_{p,s,c,1} < \mu^y_{p,s,c,2} < \mu^y_{p,s,c,3}, \quad \mu^x_{p,s,c,2} \in [4.5, 6], \quad |\mu^y_{p,s,c,2}| \in [2, 2.2]$$

Scale

$$\tau^x_{p,s,c,d} \sim \mathcal{N}\left(\tau^x_{g,s,c,d},\, \sigma_\tau\right), \quad \tau^x_{g,s,c,d} \sim \mathcal{N}(0,\, 2.5), \quad \tau^y_{p,s,c,d} \sim \mathcal{N}\left(\tau^y_{g,s,c,d},\, \sigma_\tau\right), \quad \tau^y_{g,s,c,d} \sim \mathcal{N}(0,\, 2.5)$$

$$\tau^x_{p,s,c,2} \in [0, 1], \quad \tau^y_{p,s,c,2} \in [0, 0.3]$$

Correlation

$$\rho_{p,s,c,d} \sim \mathcal{N}(\rho_{g,s,c,d},\, \sigma_\rho), \quad \rho_{g,s,c,d} \sim \mathcal{N}(0,\, 0.5), \quad \rho_{p,s,c,1} < \rho_{p,s,c,2} < \rho_{p,s,c,3}$$

Censoring

$$t_{p,s,c,d} \sim \mathcal{N}(t_{g,s,c,d},\, \sigma_t), \quad t_{g,s,c,d} \sim \mathcal{N}(3,\, 1), \quad t_{p,s,c,d} \in \left[0,\, t^{\max}_{p,s,c,d}\right]$$

Hyperpriors

$$\sigma_\mu \sim \mathcal{N}^+(0,\, 2.5), \quad \sigma_\tau \sim \mathcal{N}^+(0,\, 2.5), \quad \sigma_\rho \sim \mathcal{N}^+(0,\, 1), \quad \sigma_t \sim \mathcal{N}^+(0,\, 1)$$

Validation

$$W_i \sim \text{Binomial}(n_i,\, p_i)$$

$$\text{logit}(p_i) = \alpha_{p(i)} + \beta_{q(i)} + \gamma_{r(i)} + \delta \cdot s_i + \left(\eta + \zeta_{r(i)} + \xi \cdot s_i\right) d_i$$